## DIAGNOSTIC STUDIES

Laboratory studies for initial evaluation should include a complete blood count, coagulation studies, basic metabolic panel, and urinalysis (to evaluate for pulmonary-renal disease). The absence of anemia in life-threatening hemoptysis is not uncommon, as asphyxiation can occur without large blood volume loss. Infectious workup (including bacterial sputum culture, acid-fast bacillus culture, and respiratory viral panel) and serologic workup (including antinuclear antibody, antineutrophilic cytoplasmic antibody, and anti-GBM) can be obtained if indicated.

Imaging of the chest is essential. While a chest radiograph is often first obtained, aside from large masses, it often does not localize the site and can be normal. In most patients without risk factors for malignancy or other abnormalities in the initial evaluation and with a normal chest radiograph, treating for bronchitis and ensuring close follow-up is a reasonable strategy, with further diagnostic workup.

In contrast, patients with risk factors for malignancy (i.e., age >40, especially those with risk factors for malignancy or significant hemoptysis) should have a chest computed tomography (CT). A chest CT identifies the site and cause of hemoptysis in the majority of patients. A CT angiogram can be considered to identify a source for possible embolization if needed. A diagnostic bronchoscopy can be a complimentary study performed to evaluate for subtle mucosal lesions that are often missed on CT scans, perform interventions as per below, or localize the general location and laterality of bleeding. This is best performed during an acute episode of hemoptysis to increase the yield.

## INTERVENTIONS

When the amount of hemoptysis is life-threatening, there are three simultaneous goals: first, protect the nonbleeding lung; second, locate the site of bleeding; and third, control the bleeding.

Protecting the airway and nonbleeding lung is paramount in the management of life-threatening hemoptysis because asphyxiation can happen quickly. If the side of bleeding is known, the patient should be positioned with the bleeding side down to use gravitational advantage to keep blood out of the nonbleeding lung. A strong cough reflex can often be highly effective at clearing clot from the airway, but intubation is sometimes necessary if the patient cannot protect their airway. Intubation with a large endotracheal tube (size 8.5) is preferred to allow for suctioning and interventions. If needed, this can be advanced into the mainstem airway of the nonbleeding lung for isolation. Double-lumen endotracheal tubes can also be used, but the lumen of each is often too small for suctioning and therapeutic interventions.

Imaging obtained during initial evaluation can indicate the source of the bleeding. Flexible bronchoscopy during active bleeding is most useful. Even if a bleeding lesion is not identified, flexible bronchoscopy can help identify a side and lobe. Finally, proceeding directly to CT angiography is also a reasonable strategy given that it has both diagnostic and therapeutic capabilities.

Controlling the bleeding during an episode of life-threatening hemoptysis can be accomplished in one of three ways: from the airway lumen, from the involved blood vessel, or by surgical resection of both airway and vessel involved. Any known bleeding diathesis should be corrected first. As above, bronchoscopic measures can often be diagnostic, but also therapeutic if the bleeding is from a central airway lesion. Initially, a flexible bronchoscope can be used to help stabilize a patient via suctioning clot and insertion of a balloon catheter or bronchial blocker to occlude the involved airway. Additionally, instillation of iced saline or epinephrine can help temporize bleeding. Tranexamic acid (TXA), an antifibrinolytic agent (500 mg/5 mL), can be instilled during bronchoscopy or administered via nebulizer. Rigid bronchoscopy, done by an interventional pulmonologist or thoracic surgeon, allows therapeutic interventions of bleeding airway lesions such as argon plasma coagulation and cautery, which can be definitive. However, because most life-threatening cases of hemoptysis arise from the bronchial circulation, bronchial artery embolization is the procedure of choice for control of the bleeding. It is generally successful in the short term, with >80% success rate at controlling bleeding immediately, although bleeding can recur if the underlying disease (e.g., a mycetoma) is not treated. Surgical resection has a high mortality rate (up to 15–40%) and should not be pursued unless initial measures have failed and bleeding is ongoing. Ideal candidates for surgery have localized disease but otherwise normal lung parenchyma.

**Acknowledgment**

*Anna K. Brady and Patricia A. Kritek contributed to this chapter in the 20th edition and some material from that chapter has been retained here.*

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# 42 Hypoxia and Cyanosis

**Joseph Loscalzo**

## HYPOXIA

The fundamental purpose of the cardiorespiratory system is to deliver $O_2$ and nutrients to cells and to remove $CO_2$ and other metabolic products from them. Proper maintenance of this function depends not only on intact cardiovascular and respiratory systems, but also on an adequate number of red blood cells and hemoglobin, and a supply of inspired gas containing adequate $O_2$.

## RESPONSES TO HYPOXIA

Decreased $O_2$ availability to cells typically results in an inhibition of oxidative phosphorylation and increased anaerobic glycolysis. This switch from aerobic to anaerobic metabolism, the Pasteur effect, reduces the yield of adenosine 5′-triphosphate (ATP) produced per mole of glucose. In severe hypoxia, when ATP production is inadequate to meet the energy requirements of ionic and osmotic equilibrium, cell membrane depolarization leads to uncontrolled $Ca^{2+}$ influx and activation of $Ca^{2+}$-dependent phospholipases and proteases. These events, in turn, cause cell swelling, activation of apoptotic pathways, and, ultimately, cell death.

The adaptations to hypoxia are mediated, in part, by the upregulation of genes encoding a variety of proteins, including glycolytic enzymes, such as phosphoglycerate kinase and phosphofructokinase, as well as the glucose transporters Glut-1 and Glut-2; and by growth factors, such as vascular endothelial growth factor (VEGF) and erythropoietin, which enhance erythrocyte production. The hypoxia-induced increase in expression of these and other key proteins is largely governed by the hypoxia-sensitive transcription factor, hypoxia-inducible factor-1 (HIF-1).

During hypoxia, systemic arterioles dilate, at least in part, by opening of $K_{ATP}$ channels in vascular smooth-muscle cells due to the hypoxia-induced reduction in ATP concentration. By contrast, in pulmonary

vascular smooth-muscle cells, inhibition of $K^+$ channels causes depolarization, which, in turn, activates voltage-gated $Ca^{2+}$ channels, raising the cytosolic [$Ca^{2+}$] and causing smooth-muscle cell contraction. Hypoxia-induced pulmonary arterial constriction shunts blood away from poorly ventilated portions toward better ventilated portions of the lung (i.e., improves ventilation-perfusion mismatch); however, it also increases pulmonary vascular resistance and right ventricular afterload.

**Effects on the Central Nervous System** Changes in the central nervous system (CNS) function, particularly the higher centers, are especially important consequences of hypoxia. Acute hypoxia causes impaired judgment, motor incoordination, and a clinical picture resembling acute alcohol intoxication. High-altitude illness is characterized by headache secondary to cerebral vasodilation, gastrointestinal symptoms, dizziness, insomnia, fatigue, or somnolence. Pulmonary arterial and sometimes venous constriction causes capillary leakage and high-altitude pulmonary edema (HAPE) **(Chap. 39)**, which intensifies hypoxia, further promoting vasoconstriction. Rarely, high-altitude cerebral edema (HACE) develops, which is manifest by severe headache and papilledema, and can cause coma. As hypoxia becomes more severe, the regulatory centers of the brainstem are affected, and death usually results from respiratory failure.

**Effects on the Cardiovascular System** Acute hypoxia stimulates the chemoreceptor reflex arc to induce venoconstriction and systemic arterial vasodilation. These acute changes are accompanied by transiently increased myocardial contractility, which is followed by depressed myocardial contractility with prolonged hypoxia.

## CAUSES OF HYPOXIA

**Respiratory Hypoxia** When hypoxia occurs from respiratory failure, $Pa_{O_2}$ declines, and when respiratory failure is persistent, the hemoglobin-oxygen (Hb-$O_2$) dissociation curve **(see Fig. 103-2)** is displaced to the right, with greater quantities of $O_2$ released at any level of tissue $P_{O_2}$. Arterial hypoxemia, that is, a reduction of $O_2$ saturation of arterial blood ($Sa_{O_2}$), and consequent cyanosis are likely to be more marked when such depression of $Pa_{O_2}$ results from pulmonary disease than when the depression occurs as the result of a decline in the fraction of oxygen in inspired air ($F_{IO_2}$). In this latter situation, $Pa_{CO_2}$ falls secondary to anoxia-induced hyperventilation and the Hb-$O_2$ dissociation curve is displaced to the left, limiting the decline in $Sa_{O_2}$ at any level of $Pa_{O_2}$.

The most common cause of respiratory hypoxia is *ventilation-perfusion mismatch* resulting from perfusion of poorly ventilated alveoli. Respiratory hypoxemia may also be caused by *hypoventilation*, in which case it is associated with an elevation of $Pa_{CO_2}$ **(Chap. 296)**. These two forms of respiratory hypoxia are usually correctable by inspiring 100% $O_2$ for several minutes. A third cause of respiratory hypoxia is shunting of blood across the lung from the pulmonary arterial to the venous bed (*intrapulmonary right-to-left shunting*) by perfusion of nonventilated portions of the lung, as in pulmonary atelectasis or through pulmonary arteriovenous connections. The low $Pa_{O_2}$ in this situation is only partially corrected by an $F_{IO_2}$ of 100%.

**Hypoxia Secondary to High Altitude** As one ascends rapidly to 3000 m (~10,000 ft), the reduction of the $O_2$ content of inspired air ($F_{IO_2}$) leads to a decrease in alveolar $P_{O_2}$ to ~60 mmHg, and a condition termed *high-altitude illness* develops (see above). At higher altitudes, arterial saturation declines rapidly and symptoms become more severe; and at 5000 m, unacclimated individuals usually cease to be able to function normally owing to the changes in CNS function described above.

**Hypoxia Secondary to Right-to-Left Extrapulmonary Shunting** From a physiologic viewpoint, this cause of hypoxia resembles intrapulmonary right-to-left shunting but is caused by congenital cardiac malformations, such as tetralogy of Fallot, transposition of the great arteries, atrial or ventricular septal defect, patent ductus arteriosus, and Eisenmenger's syndrome **(Chap. 280)**. As in pulmonary right-to-left shunting, the $Pa_{O_2}$ cannot be restored to normal with inspiration of 100% $O_2$.

**Anemic Hypoxia** A reduction in hemoglobin concentration of the blood is accompanied by a corresponding decline in the $O_2$-carrying capacity of the blood. Although the $Pa_{O_2}$ is normal in anemic hypoxia, the absolute quantity of $O_2$ transported per unit volume of blood is diminished. As the anemic blood passes through the capillaries and the usual quantity of $O_2$ is removed from it, the $P_{O_2}$ and saturation in the venous blood decline to a greater extent than normal.

**Carbon Monoxide (CO) Intoxication** **(See also Chap. 476)** Hemoglobin that binds CO (carboxy-hemoglobin [COHb]) is unavailable for $O_2$ transport. In addition, the presence of COHb shifts the Hb-$O_2$ dissociation curve to the left **(see Fig. 103-2)** so that $O_2$ is unloaded only at lower tensions, further contributing to tissue hypoxia.

**Circulatory Hypoxia** As in anemic hypoxia, the $Pa_{O_2}$ is usually normal, but venous and tissue $P_{O_2}$ values are reduced as a consequence of reduced tissue perfusion and greater tissue $O_2$ extraction. This pathophysiology leads to an increased arterial-mixed venous $O_2$ difference (a-v-$O_2$ difference), or gradient. Generalized circulatory hypoxia occurs in heart failure **(Chap. 264)** and in most forms of shock **(Chap. 314)**.

**Specific Organ Hypoxia** Localized circulatory hypoxia may occur as a result of decreased perfusion secondary to arterial obstruction, as in localized atherosclerosis in any vascular bed, or as a consequence of vasoconstriction, as observed in Raynaud's phenomenon **(Chap. 292)**. Localized hypoxia may also result from venous obstruction and the resultant expansion of interstitial fluid causing arteriolar compression and, thereby, reduction of arterial inflow. Edema, which increases the distance through which $O_2$ must diffuse before it reaches cells, can also cause localized hypoxia. In an attempt to maintain adequate perfusion to more vital organs in patients with reduced cardiac output secondary to heart failure or hypovolemic shock, vasoconstriction may reduce perfusion in the limbs and skin, causing hypoxia of these regions.

**Increased $O_2$ Requirements** If the $O_2$ consumption of tissues is elevated without a corresponding increase in perfusion, tissue hypoxia ensues and the $P_{O_2}$ in venous blood declines. Ordinarily, the clinical picture of patients with hypoxia due to an elevated metabolic rate, as in fever or thyrotoxicosis, is quite different from that in other types of hypoxia: the skin is warm and flushed owing to increased cutaneous blood flow that dissipates the excessive heat produced, and cyanosis is usually absent.

Exercise is a classic example of increased tissue $O_2$ requirements. These increased demands are normally met by several mechanisms operating simultaneously: (1) an increase in the cardiac output and ventilation and, thus, $O_2$ delivery to the tissues; (2) a preferential shift in blood flow to the exercising muscles by changing vascular resistances in the circulatory beds of exercising tissues, directly and/or reflexly; (3) an increase in $O_2$ extraction from the delivered blood and a widening of the arteriovenous $O_2$ difference; and (4) a reduction in the pH of the tissues and capillary blood, shifting the Hb-$O_2$ curve to the right **(see Fig. 103-2)** and thereby unloading more $O_2$ from hemoglobin. If the capacity of these mechanisms is exceeded, then hypoxia, especially of the exercising muscles, will result.

**Improper Oxygen Utilization** Cyanide **(Chap. 470)** and several other similarly acting poisons cause cellular hypoxia by impairing electron transport in mitochondria, thereby limiting oxidative phosphorylation and ATP production. The tissues are unable to use $O_2$, and as a consequence, the venous blood tends to have a high $O_2$ tension. This condition has been termed *histotoxic hypoxia*.

## ADAPTATION TO HYPOXIA

An important component of the respiratory response to hypoxia originates in special chemosensitive cells in the carotid and aortic bodies and in the respiratory center in the brainstem. The stimulation of these cells by hypoxia increases ventilation, with a loss of $CO_2$, and can lead to respiratory alkalosis. When combined with the metabolic acidosis resulting from the production of lactic acid, the serum bicarbonate level declines **(Chap. 58)**.

With the reduction of $Pa_{O_2}$, cerebrovascular resistance decreases and cerebral blood flow increases in an attempt to maintain $O_2$ delivery to the brain. However, when the reduction of $Pa_{O_2}$ is accompanied by hyperventilation and a reduction of $Pa_{CO_2}$, cerebrovascular resistance rises, cerebral blood flow falls, and tissue hypoxia intensifies.

The diffuse, systemic vasodilation that occurs in generalized hypoxia increases the cardiac output. In patients with underlying heart disease, the requirements of peripheral tissues for an increase of cardiac output with hypoxia may precipitate congestive heart failure. In patients with ischemic heart disease, a reduced $Pa_{O_2}$ may intensify myocardial ischemia and further impair left ventricular function.

One of the important compensatory mechanisms for chronic hypoxia is an increase in the hemoglobin concentration and in the number of red blood cells in the circulating blood, that is, the development of polycythemia induced by erythropoietin production **(Chap. 108)**. In persons with chronic hypoxemia secondary to prolonged residence at a high altitude (>13,000 ft, 4200 m), a condition termed *chronic mountain sickness* develops. This disorder is characterized by a blunted respiratory drive, reduced ventilation, erythrocytosis, cyanosis, weakness, right ventricular enlargement secondary to pulmonary hypertension, and even stupor.

## CYANOSIS

*Cyanosis* refers to a bluish color of the skin and mucous membranes resulting from an increased quantity of reduced hemoglobin (i.e., deoxygenated hemoglobin) or of hemoglobin derivatives (e.g., methemoglobin or sulfhemoglobin) in the small blood vessels of those tissues. It is usually most marked in the lips, nail beds, ears, and malar eminences. Cyanosis, especially if developed recently, is more commonly detected by a family member than the patient. A cherry-colored flush, rather than cyanosis, is caused by COHb **(Chap. 470)**.

The degree of cyanosis is modified by the color of the cutaneous pigment and the thickness of the skin, as well as by the state of the cutaneous capillaries. The accurate clinical detection of the presence and degree of cyanosis is difficult, as proved by oximetric studies. In some instances, central cyanosis can be detected reliably when the $Sa_{O_2}$ has fallen to 85%; in others, particularly in dark-skinned persons, it may not be detected until it has declined to 75%. In the latter case, examination of the mucous membranes in the oral cavity and the conjunctivae rather than examination of the skin is more helpful in the detection of cyanosis.

The increase in the quantity of reduced hemoglobin in the mucocutaneous vessels that produces cyanosis may be brought about either by an increase in the quantity of venous blood as a result of dilation of the venules (including precapillary venules) or by a reduction in the $Sa_{O_2}$ in the capillary blood. In general, cyanosis becomes apparent when the concentration of reduced hemoglobin in capillary blood exceeds 40 g/L (4 g/dL).

It is the *absolute*, rather than the *relative*, quantity of reduced hemoglobin that is important in producing cyanosis. Thus, in a patient with severe anemia, the *relative* quantity of reduced hemoglobin in the venous blood may be very large when considered in relation to the total quantity of hemoglobin in the blood. However, since the concentration of the latter is markedly reduced, the *absolute* quantity of reduced hemoglobin may still be low, and therefore, patients with severe anemia and even *marked* arterial desaturation may not display cyanosis. Conversely, the higher the total hemoglobin content, the greater is the tendency toward cyanosis; thus, patients with marked polycythemia tend to be cyanotic at higher levels of $Sa_{O_2}$ than patients with normal hematocrit values. Likewise, local passive congestion, which causes an increase in the total quantity of reduced hemoglobin in the vessels in a given area, may cause cyanosis. Cyanosis is also observed when nonfunctional hemoglobin, such as methemoglobin (consequential or acquired) or sulfhemoglobin **(Chap. 103)**, is present in blood.

Cyanosis may be subdivided into central and peripheral types. In *central* cyanosis, the $Sa_{O_2}$ is reduced or an abnormal hemoglobin derivative is present, and the mucous membranes and skin are both affected. *Peripheral* cyanosis is due to a slowing of blood flow and abnormally great extraction of $O_2$ from normally saturated arterial blood; it results from vasoconstriction and diminished peripheral blood flow, such as occurs in cold exposure, shock, congestive failure, and peripheral vascular disease. Often in these conditions, the mucous membranes of the oral cavity, including the sublingual mucosa, may be spared. Clinical differentiation between central and peripheral cyanosis may not always be straightforward, and in conditions such as cardiogenic shock with pulmonary edema, there may be a mixture of both types.

### DIFFERENTIAL DIAGNOSIS

**Central Cyanosis (Table 42-1)** Decreased $Sa_{O_2}$ results from a marked reduction in the $Pa_{O_2}$. This reduction may be brought about by a decline in the $F_{IO_2}$ without sufficient compensatory alveolar hyperventilation to maintain alveolar $P_{O_2}$. Cyanosis usually becomes manifest in an ascent to an altitude of 4000 m (13,000 ft).

Seriously *impaired pulmonary function*, through perfusion of unventilated or poorly ventilated areas of the lung or alveolar hypoventilation, is a common cause of central cyanosis **(Chap. 296)**. This condition may occur acutely, as in extensive pneumonia or pulmonary edema, or chronically, with chronic pulmonary diseases (e.g., emphysema). In the latter situation, secondary polycythemia is generally present and clubbing of the fingers (see below) may occur. Another cause of reduced $Sa_{O_2}$ is *shunting of systemic venous blood into the arterial circuit*. Certain forms of congenital heart disease are associated with cyanosis on this basis (see above and **Chap. 280**).

*Pulmonary arteriovenous fistulae* may be congenital or acquired, solitary or multiple, and microscopic or massive. The severity of cyanosis produced by these fistulae depends on their size and number. They occur with some frequency in hereditary hemorrhagic telangiectasia. $Sa_{O_2}$ reduction and cyanosis may also occur in some patients with cirrhosis, presumably as a consequence of pulmonary arteriovenous fistulae or portal vein–pulmonary vein anastomoses.

In patients with cardiac or pulmonary right-to-left shunts, the presence and severity of cyanosis depend on the size of the shunt relative to the systemic flow and on the Hb-$O_2$ saturation of the venous blood. With increased extraction of $O_2$ from the blood by the exercising muscles, the venous blood returning to the right side of the heart is more unsaturated than at rest, and shunting of this blood intensifies the cyanosis. Secondary polycythemia occurs frequently in patients in this setting and contributes to the cyanosis.

**TABLE 42-1 Causes of Cyanosis**

| |
|---|
| **Central Cyanosis** |
| Decreased arterial oxygen saturation |
| Decreased atmospheric pressure—high altitude |
| Impaired pulmonary function |
| Alveolar hypoventilation |
| Inhomogeneity in pulmonary ventilation and perfusion (perfusion of hypoventilated alveoli) |
| Impaired oxygen diffusion |
| Anatomic shunts |
| Certain types of congenital heart disease |
| Pulmonary arteriovenous fistulas |
| Multiple small intrapulmonary shunts |
| Hemoglobin with low affinity for oxygen |
| Hemoglobin abnormalities |
| Methemoglobinemia—hereditary, acquired |
| Sulfhemoglobinemia—acquired |
| Carboxyhemoglobinemia (not true cyanosis) |
| **Peripheral Cyanosis** |
| Reduced cardiac output |
| Cold exposure |
| Redistribution of blood flow from extremities |
| Arterial obstruction |
| Venous obstruction |

Cyanosis can be caused by small quantities of circulating methemoglobin (Hb $Fe^{3+}$) and by even smaller quantities of sulfhemoglobin (Chap. 103); both of these hemoglobin derivatives impair oxygen delivery to the tissues. Although they are uncommon causes of cyanosis, these abnormal hemoglobin species should be sought by spectroscopy when cyanosis is not readily explained by malfunction of the circulatory or respiratory systems. Generally, digital clubbing does not occur with these conditions.

**Peripheral Cyanosis** Probably the most common cause of peripheral cyanosis is the normal vasoconstriction resulting from exposure to cold air or water. When cardiac output is reduced, cutaneous vasoconstriction occurs as a compensatory mechanism so that blood is diverted from the skin to more vital areas such as the CNS and heart, and cyanosis of the extremities may result even though the arterial blood is normally saturated.

Arterial obstruction to an extremity, as with an embolus, or arteriolar constriction, as in cold-induced vasospasm (Raynaud's phenomenon) (Chap. 292), generally results in pallor and coldness, and there may be associated cyanosis. Venous obstruction, as in thrombophlebitis or deep venous thrombosis, dilates the subpapillary venous plexuses and thereby intensifies cyanosis.

## APPROACH TO THE PATIENT

### Cyanosis

Certain features are important in arriving at the cause of cyanosis:

1. It is important to ascertain the time of onset of cyanosis. Cyanosis present since birth or infancy is usually due to congenital heart disease.
2. Central and peripheral cyanosis must be differentiated. Evidence of disorders of the respiratory or cardiovascular systems is helpful. Massage or gentle warming of a cyanotic extremity will increase peripheral blood flow and abolish peripheral, but not central, cyanosis.
3. The presence or absence of clubbing of the digits (see below) should be ascertained. The combination of cyanosis and clubbing is frequent in patients with congenital heart disease and right-to-left shunting and is seen occasionally in patients with pulmonary disease, such as lung abscess or pulmonary arteriovenous fistulae. In contrast, peripheral cyanosis or acutely developing central cyanosis is *not* associated with clubbed digits.
4. $Pa_{O_2}$ and $Sa_{O_2}$ should be determined, and in patients with cyanosis in whom the mechanism is obscure, spectroscopic examination of the blood should be performed to look for abnormal types of hemoglobin (critical in the differential diagnosis of cyanosis).

## CLUBBING

The selective bulbous enlargement of the distal segments of the fingers and toes due to proliferation of connective tissue, particularly on the dorsal surface, is termed *clubbing*; there is also increased sponginess of the soft tissue at the base of the clubbed nail. Clubbing may be hereditary, idiopathic, or acquired and associated with a variety of disorders, including cyanotic congenital heart disease (see above), infective endocarditis, and a variety of pulmonary conditions (among them primary and metastatic lung cancer, bronchiectasis, asbestosis, sarcoidosis, lung abscess, cystic fibrosis, tuberculosis, and mesothelioma), as well as with some gastrointestinal diseases (including inflammatory bowel disease and hepatic cirrhosis). In some instances, it is occupational, for example, in jackhammer operators.

Clubbing in patients with primary and metastatic lung cancer, mesothelioma, bronchiectasis, or hepatic cirrhosis may be associated with *hypertrophic osteoarthropathy*. In this condition, the subperiosteal formation of new bone in the distal diaphyses of the long bones of the extremities causes pain and symmetric arthritis-like changes in the shoulders, knees, ankles, wrists, and elbows. The diagnosis of hypertrophic osteoarthropathy may be confirmed by bone radiograph or magnetic resonance imaging (MRI). Although the mechanism of clubbing is unclear, it appears to be secondary to humoral substances that cause dilation of the vessels of the distal digits as well as growth factors released from platelet precursors in the digital circulation. In certain circumstances, clubbing is reversible, such as following lung transplantation for cystic fibrosis.

### FURTHER READING


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# 43 Edema

Joseph Loscalzo

## PLASMA AND INTERSTITIAL FLUID EXCHANGE

Approximately two-thirds of total body water is intracellular and one-third is extracellular. One-fourth of the latter is in the plasma, and the remainder comprises the interstitial fluid. Edema represents an excess of interstitial fluid that has become evident clinically.

There is constant interchange of fluid between the two compartments of the extracellular fluid. The hydrostatic pressure within the capillaries and the colloid oncotic pressure in the interstitial fluid promote the movement of water and diffusible solutes from plasma to the interstitium. This movement is most prominent at the arterial origin of the capillary and falls progressively with the decline in intracapillary pressure and the rise in oncotic pressure toward the venular end. Fluid is returned from the interstitial space into the vascular system largely through the lymphatic system. These interchanges of fluids are normally balanced so that the volumes of the intravascular and interstitial compartments remain constant. However, a net movement of fluid from the intravascular to the interstitial spaces takes place and may be responsible for the development of edema under the following conditions: (1) an increase in intracapillary hydrostatic pressure; (2) inadequate lymphatic drainage; (3) reductions in the oncotic pressure of plasma; (4) damage to or dysfunction of the capillary endothelial barrier; and (5) increases in the oncotic pressure of the interstitial space.

### REDUCTION OF EFFECTIVE ARTERIAL VOLUME

In many forms of edema, the effective arterial blood volume, a parameter that represents the filling of the arterial tree and that effectively perfuses the tissues, is reduced. Underfilling of the arterial tree may be caused by a reduction of cardiac output and/or systemic vascular resistance, by the pooling of blood in the splanchnic veins (as in cirrhosis), and by hypoalbuminemia (Fig. 43-1*A*). As a consequence of this underfilling, a series of physiological responses designed to restore the effective arterial volume to normal are set into motion. A key element of these responses is the renal retention of sodium and, therefore, water, thereby restoring effective arterial volume, but sometimes also leading to the development or intensification of edema.

### RENAL FACTORS AND THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

The diminished renal blood flow characteristic of states in which the effective arterial blood volume is reduced is translated by the renal